P7 OPERATING SYSTEM - Process Management

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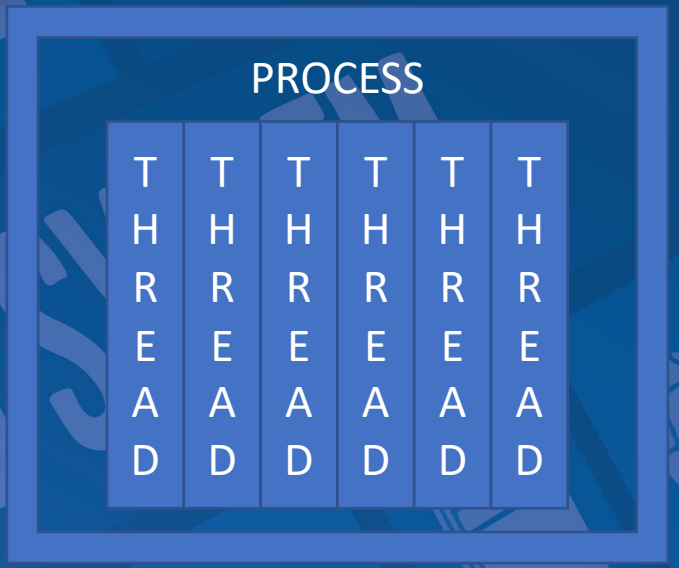
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Process Management (Processes and Threads)

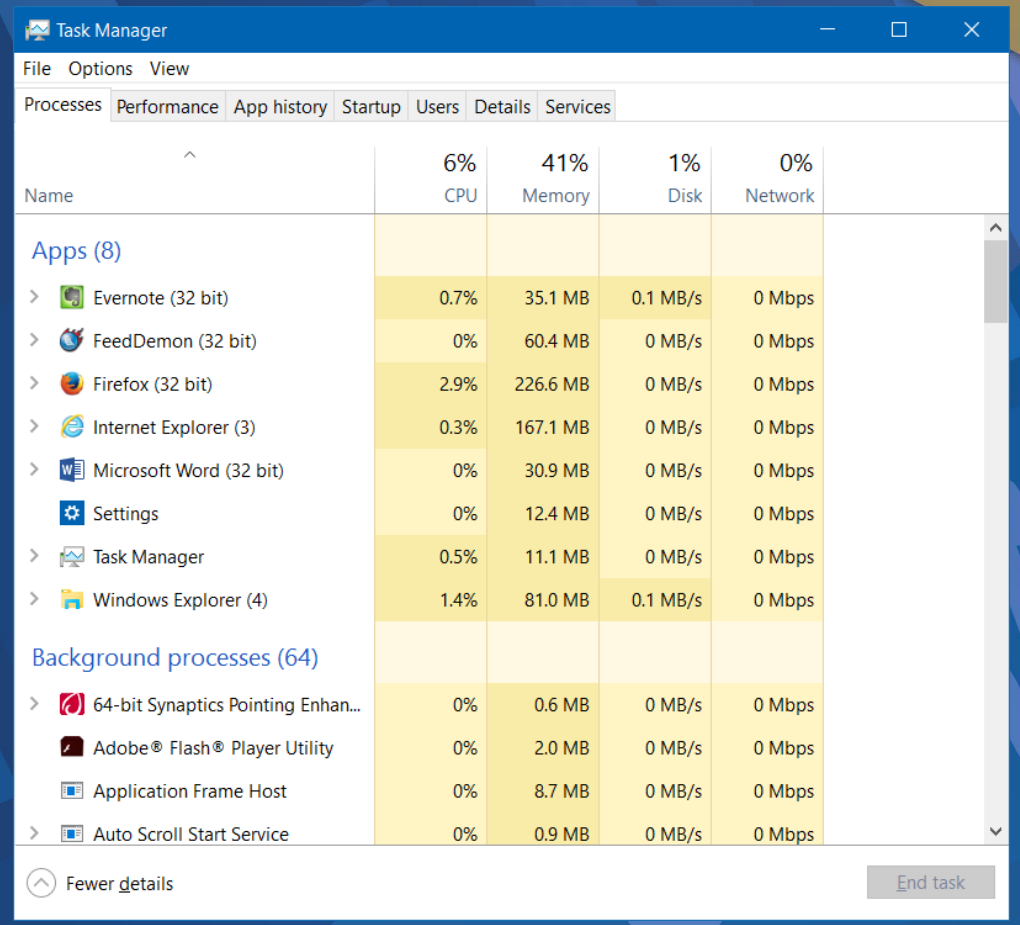


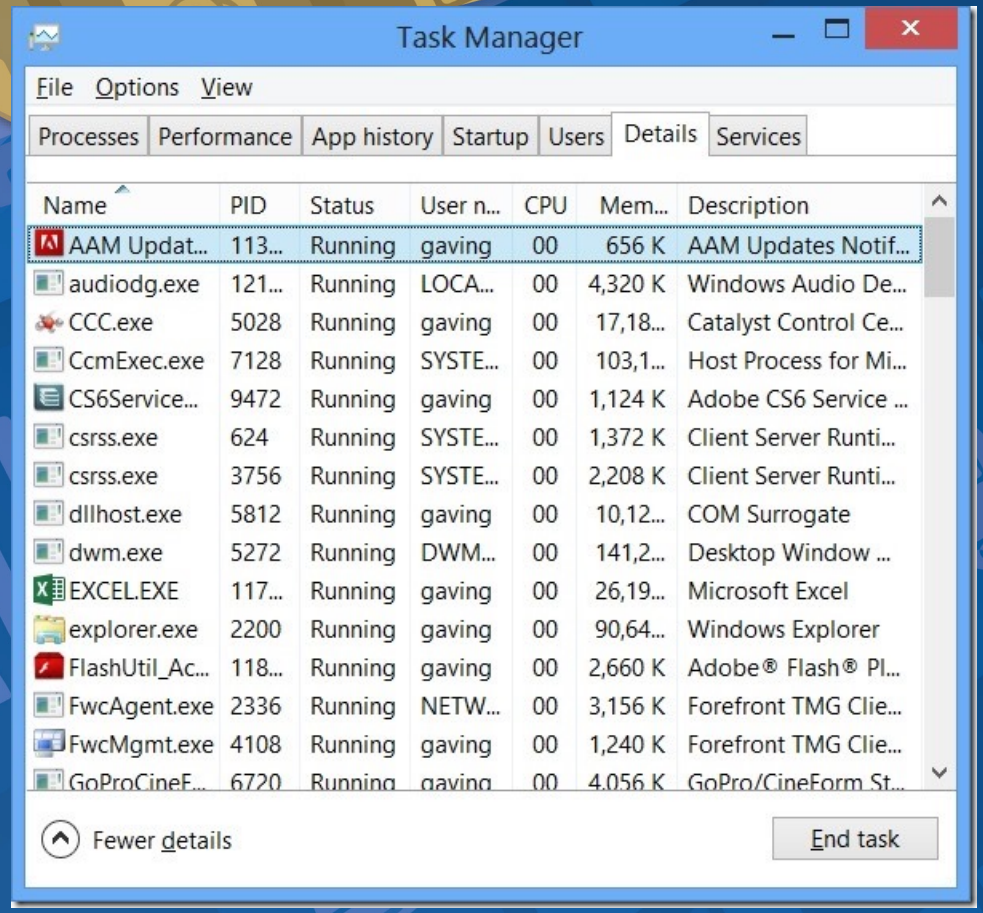
Process :

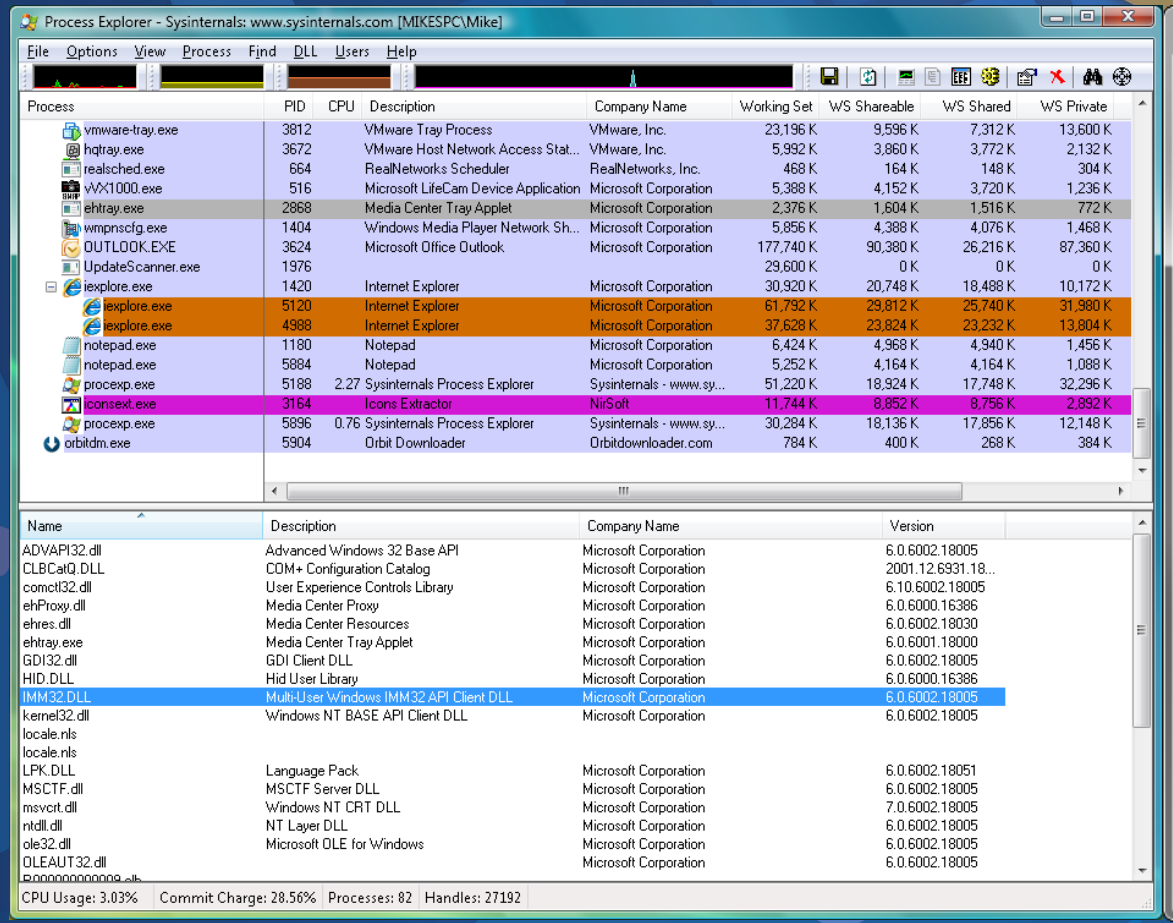
- A process can be thought of as a program in execution

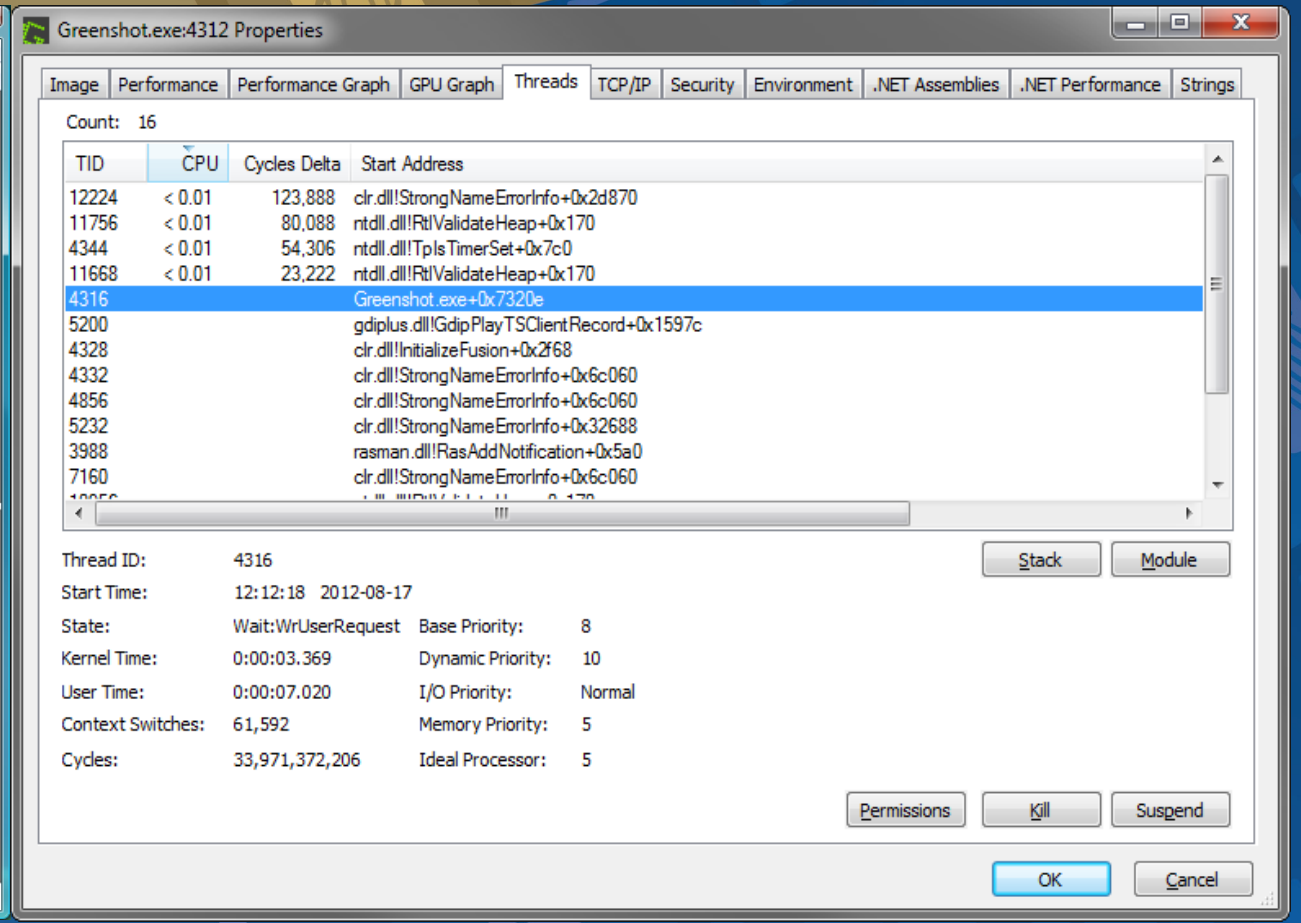
Thread :

- A thread is the unit of execution within a process. A process can have anywhere from just one thread to many threads

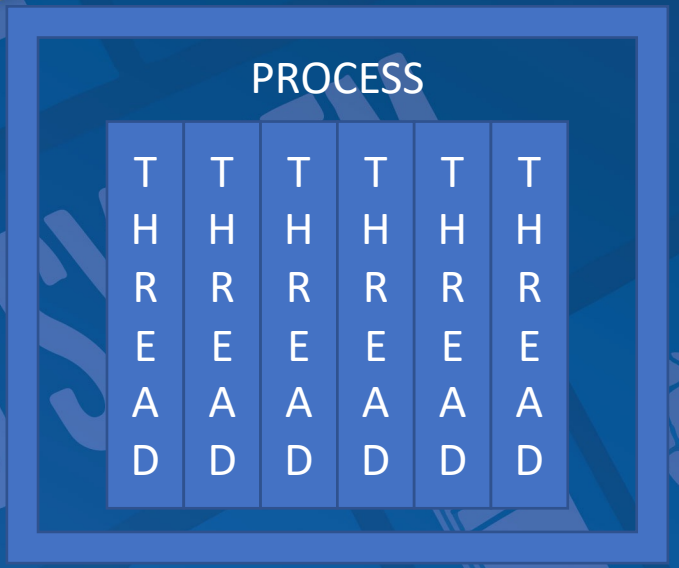

Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Name
6% CPU
41% Memory
1% Disk
0% Network
Apps (8)
Evernote (32 bit) 0.7% 35.1 MB 0.1 MB/s 0 Mbps
FeedDemon (32 bit) 0% 60.4 MB 0 MB/s 0 Mbps
Firefox (32 bit) 2.9% 226.6 MB 0 MB/s 0 Mbps
Internet Explorer (3) 0.3% 167.1 MB 0 MB/s 0 Mbps
Microsoft Word (32 bit) 0% 30.9 MB 0 MB/s 0 Mbps
Settings 0% 12.4 MB 0 MB/s 0 Mbps
Task Manager 0.5% 11.1 MB 0 MB/s 0 Mbps
Windows Explorer (4) 1.4% 81.0 MB 0.1 MB/s 0 Mbps
Background processes (64)
64-bit Synaptics Pointing Enhan... 0% 0.6 MB 0 MB/s 0 Mbps
Adobe® Flash® Player Utility 0% 2.0 MB 0 MB/s 0 Mbps
Application Frame Host 0% 8.7 MB 0 MB/s 0 Mbps
Auto Scroll Start Service 0% 0.9 MB 0 MB/s 0 Mbps
Fewer details
End task


Task Manager
File Options View
Processes Performance App history Startup Users Details Services
Name PID Status User n... CPU Mem... Description
AAM Updat... 113... Running gaving 00 656 K AAM Updates Notif...
audiodg.exe 121... Running LOCA... 00 4,320 K Windows Audio De...
CCC.exe 5028 Running gaving 00 17,18... Catalyst Control Ce...
CcmExec.exe 7128 Running SYSTE... 00 103,1... Host Process for Mi...
CS6Service... 9472 Running gaving 00 1,124 K Adobe CS6 Service ...
csrss.exe 624 Running SYSTE... 00 1,372 K Client Server Runti...
csrss.exe 3756 Running SYSTE... 00 2,208 K Client Server Runti...
dllhost.exe 5812 Running gaving 00 10,12... COM Surrogate
dwm.exe 5272 Running DWM... 00 141,2... Desktop Window ...
EXCEL.EXE 117... Running gaving 00 26,19... Microsoft Excel
explorer.exe 2200 Running gaving 00 90,64... Windows Explorer
FlashUtil_Ac... 118... Running gaving 00 2,660 K Adobe® Flash® Pl...
FwcAgent.exe 2336 Running NETW... 00 3,156 K Forefront TMG Clie...
FwcMgmt.exe 4108 Running gaving 00 1,240 K Forefront TMG Clie...
Fewer details
End task


Process Explorer - Sysinternals: www.sysinternals.com [MIKESPC\Mike]
File Options View Process Find DLL Users Help
Process PID CPU Description Company Name Working Set WS Shareable WS Shared WS Private
vmware-tray.exe 3812 VMware Tray Process VMware, Inc. 23,196 K 9,596 K 7,312 K 13,600 K
hqtray.exe 3672 VMware Host Network Access Stat... VMware, Inc. 5,992 K 3,860 K 3,772 K 2,132 K
realsched.exe 664 RealNetworks Scheduler RealNetworks, Inc. 468 K 164 K 148 K 304 K
vVX1000.exe 516 Microsoft LifeCam Device Application Microsoft Corporation 5,388 K 4,152 K 3,720 K 1,236 K
ehtray.exe 2868 Media Center Tray Applet Microsoft Corporation 2,376 K 1,604 K 1,516 K 772 K
wmpnscfg.exe 1404 Windows Media Player Network Sh... Microsoft Corporation 5,856 K 4,388 K 4,076 K 1,468 K
OUTLOOK.EXE 3624 Microsoft Office Outlook Microsoft Corporation 177,740 K 90,380 K 26,216 K 87,360 K
UpdateScanner.exe 1976 29,600 K 0 K 0 K 0 K
iexplore.exe 1420 Internet Explorer Microsoft Corporation 30,920 K 20,748 K 18,488 K 10,172 K
iexplore.exe 5120 Internet Explorer Microsoft Corporation 61,792 K 29,812 K 25,740 K 31,980 K
iexplore.exe 4988 Internet Explorer Microsoft Corporation 37,628 K 23,824 K 23,232 K 13,804 K
notepad.exe 1180 Notepad Microsoft Corporation 6,424 K 4,968 K 4,940 K 1,456 K
notepad.exe 5884 Notepad Microsoft Corporation 5,252 K 4,164 K 4,164 K 1,088 K
procexp.exe 5188 2.27 Sysinternals Process Explorer Sysinternals - www.sy... 51,220 K 18,924 K 17,748 K 32,296 K
iconext.exe 3164 Icons Extractor NirSoft 11,744 K 8,852 K 8,756 K 2,892 K
procexp.exe 5896 0.76 Sysinternals Process Explorer Sysinternals - www.sy... 30,284 K 18,136 K 17,856 K 12,148 K
orbitdm.exe 5904 Orbit Downloader Orbitdownloader.com 784 K 400 K 268 K 384 K
Name Description Company Name Version
ADVAPI32.dll Advanced Windows 32 Base API Microsoft Corporation 6.0.6002.18005
CLBCatQ.DLL COM+ Configuration Catalog Microsoft Corporation 2001.12.6931.18...
comctl32.dll User Experience Controls Library Microsoft Corporation 6.10.6002.18005
ehProxy.dll Media Center Proxy Microsoft Corporation 6.0.6000.16386
ehres.dll Media Center Resources Microsoft Corporation 6.0.6002.18030
ehtray.exe Media Center Tray Applet Microsoft Corporation 6.0.6001.18000
GDI32.dll GDI Client DLL Microsoft Corporation 6.0.6002.18005
HID.DLL Hid User Library Microsoft Corporation 6.0.6000.16386
IMM32.DLL Multi-User Windows IMM32 API Client DLL Microsoft Corporation 6.0.6002.18005
kernel32.dll Windows NT BASE API Client DLL Microsoft Corporation 6.0.6002.18005
locale.nls
locale.nls
LPK.DLL Language Pack Microsoft Corporation 6.0.6002.18051
MSCTF.dll MSCTF Server DLL Microsoft Corporation 6.0.6002.18005
msvcrt.dll Windows NT CRT DLL Microsoft Corporation 7.0.6002.18005
ntdll.dll NT Layer DLL Microsoft Corporation 6.0.6002.18005
ole32.dll Microsoft OLE for Windows Microsoft Corporation 6.0.6002.18005
OLEAUT32.dll Microsoft Corporation 6.0.6002.18005
CPU Usage: 3.03% Commit Charge: 28.56% Processes: 82 Handles: 27192


Greenshot.exe:4312 Properties
Image Performance Performance Graph GPU Graph Threads TCP/IP Security Environment .NET Assemblies .NET Performance Strings
Count: 16
TID CPU Cycles Delta Start Address
12224 < 0.01 123,888 clr.dll!StrongNameErrorInfo+0x2d870
11756 < 0.01 80,088 ntdll.dll!RtlValidateHeap+0x170
4344 < 0.01 54,306 ntdll.dll!TpIsTimerSet+0x7c0
11668 < 0.01 23,222 ntdll.dll!RtlValidateHeap+0x170
4316 Greenshot.exe+0x7320e
5200 gdiplus.dll!GdipPlayTSClientRecord+0x1597c
4328 clr.dll!InitializeFusion+0x2f68
4332 clr.dll!StrongNameErrorInfo+0x6c060
4856 clr.dll!StrongNameErrorInfo+0x6c060
5232 clr.dll!StrongNameErrorInfo+0x32688
3988 rasman.dll!RasAddNotification+0x5a0
7160 clr.dll!StrongNameErrorInfo+0x6c060
Thread ID: 4316
Start Time: 12:12:18 2012-08-17
State: Wait:WrUserRequest
Base Priority: 8
Kernel Time: 0:00:03.369
Dynamic Priority: 10
User Time: 0:00:07.020
I/O Priority: Normal
Context Switches: 61,592
Memory Priority: 5
Cycles: 33,971,372,206
Ideal Processor: 5
Stack Module
Permissions Kill Suspend
OK Cancel

Process Management (Processes and Threads)



Process :

- A process can be thought of as a program in execution

Thread :

- A thread is the unit of execution within a process. A process can have anywhere from just one thread to many threads

Process State

- As a process executes, it changes state
- The state of a process is defined in part by the current activity of that process

Each process my be in one of the following states :

New -> The process is being created

Running -> Instructions are being executed

Waiting -> The process is waitinf for some event to occur (Such as an I/O completion or reception of a signal)

Ready -> The procces is waiting to be assigned to a processor

Terminated ->The process has finished execution

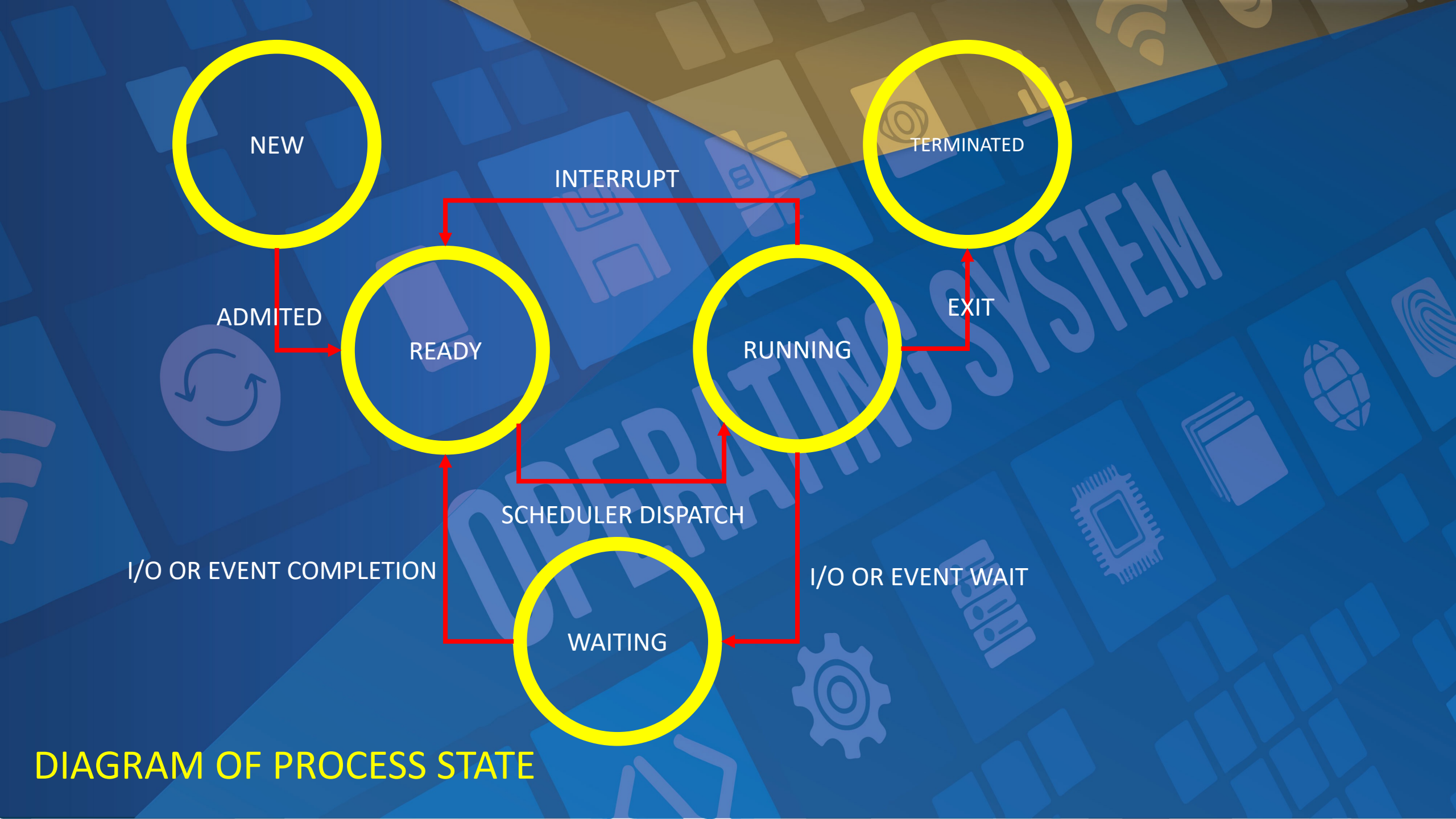
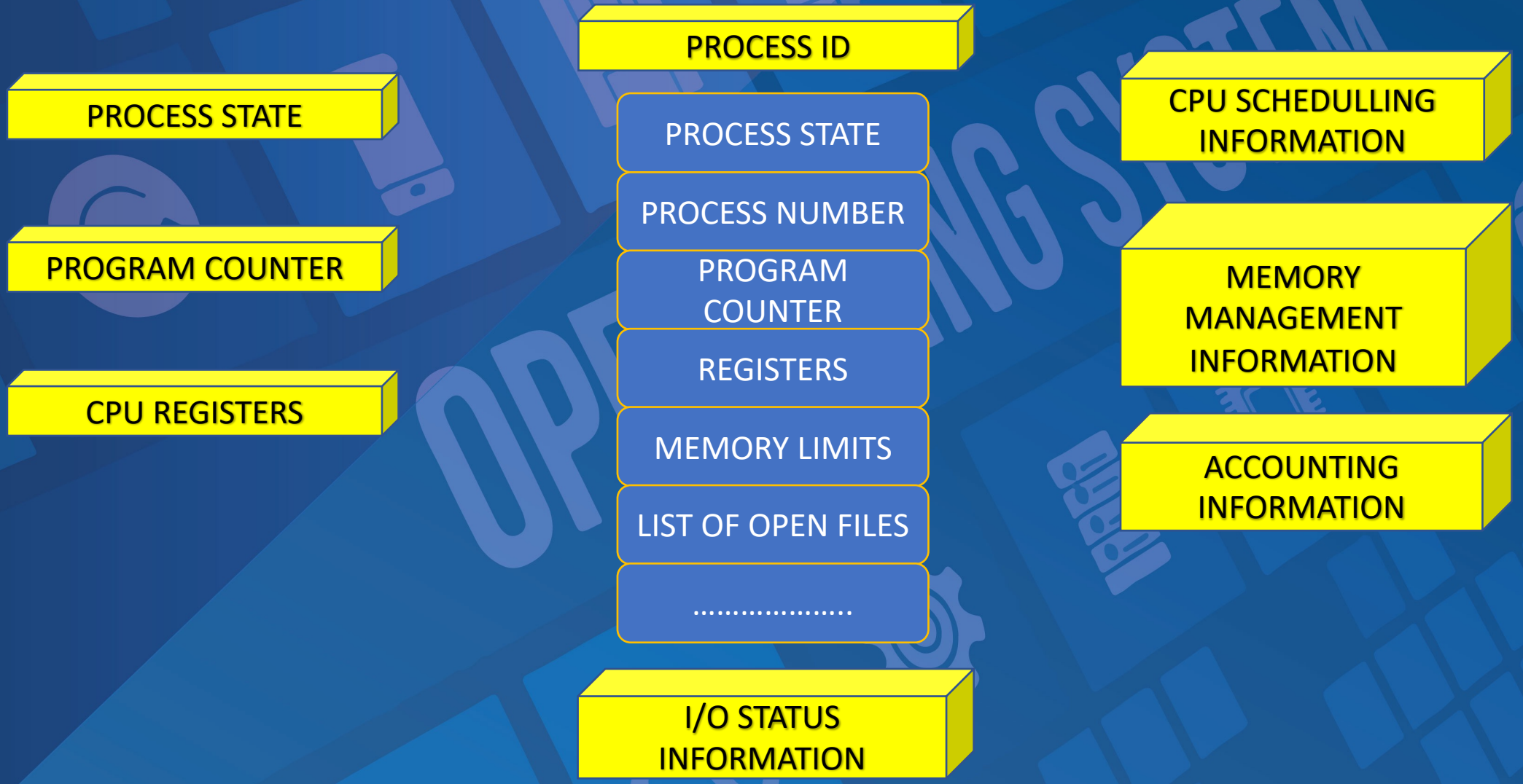


DIAGRAM OF PROCESS STATE

Process Control Block

- Each process is represented in the operating system by a Process Control Block (PCB) - also called a task control block



Process Scheduling

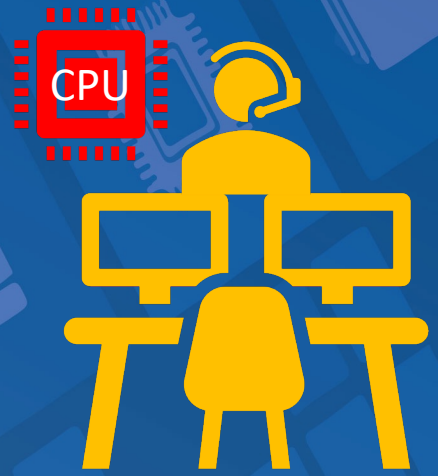
- The objective of multiprogramming is to have some process running at all times, to maxmize CPU utillization
- The objective of time sharinf is to switch the CPU among processes so frequently that users can interact with eaxh program while it is running
- To meet these objective, the process scheduler selects an available process (possibly from a set of several available processes) for program execution on the CPU
  - For a single-processor system, there will never be more than one running process
  - If there are more processes, the rest will have to wait until the CPU is free and can be rescheduled

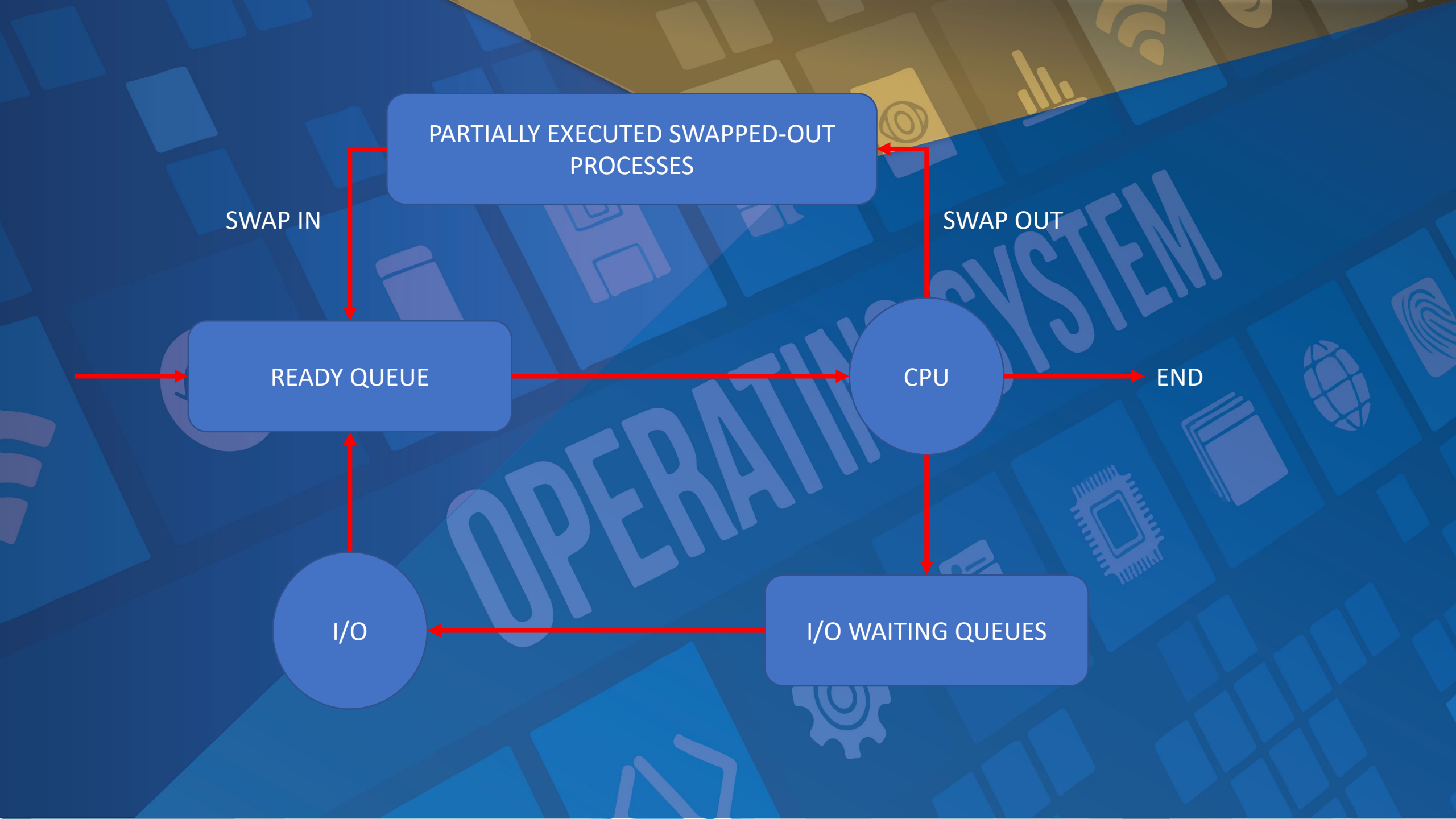
Scheduling Queues

- Job Queue : As processes enter the system, they are put into a job queue, which consists of all processes in the system
- Ready Queue : The processes that are residing in main memory and are ready and waiting to execute are kept on a list called the ready queue



PROCESS




PARTIALLY EXECUTED SWAPPED-OUT PROCESSES
SWAP IN
SWAP OUT
READY QUEUE
CPU
END
I/O
I/O WAITING QUEUES

CONTEXT SWITCH

- Interrupts cause the operating system to change a CPU from its current task and to run a kernel routine
- Such operations happen frequently on general-purpose systems

When an interrupt occurs, the system needs to save the current **context** of the process currently running on the CPU so that it can restore the context when its processing is done, essentially suspending the process and the resuming it

- The context is represented in the PCB (Process Control Block) of the process

Switching the CPU to another process requites performing a state save of the current process and a state restore of a different process.

THIS TASK IS KNOWN AS A CONTEXT SWITCH



STATE RESTORE



STATE SAVE

- Context-switch time is pure overhead, because the system does np useful work while switching
- Its speed varies from machine to machine, depending on the memory speed, the number of registers that must be copied, and the existence of special instructions (such as a single intruction to load or store all registers)
- Typical speeds are a few milliseconds

Operation on Processes (Process Creation)

- A process may creates several, via a create-procces system call, during the course of excution.
- The creting process is colled a parent process, and the new processes are colled the children of that process.
- Each of these new processes may in turn create other processes, forming a tree of processes.

a Tree of Process on a typical Solaris System



When a process create a new process, two possibilities exits in terms of rxecution :

- The parent continues to excute concurrently with its children.
- The parent waits until some or all of its children have terminated.

The are also two possibilities in therms of the address space of the new process :

- The child process is a duplicate of the parent process (it has the same program and data as the parent).
- The child process has a new program loaded into it.

Operation on Processes (Process Termination)

- A process terminates when it finishes executing its final statement and ask the operting system to delete it by using the exit () system call.
- At that point, the process may return a status value (typically an integer) to its parent process (via the wait() system call).
- All the resources of the process-including physical andvirtual memory. Open files, and I/O buffers-are deallocated by the operating system.

Termination can accur in other circumstances as well:

- A process can couse the termination of another process via an appropriate system system call.
- Usually, such a system call can be invoked only by the parent of the process that is to be termiated.
- Otherwise, users could arbitrarily kill each other's jobs.

A parent may terminate the execution of its children for a varicty of reasons, such as these :

- The child has exceeded its usage of some of the resources that it has been allocated. *(To determine whether this has accurred, the parent must have a mechanism to inspect the state of its children.)*
- The task assigned to the child is no longer required.
- The parent is exiting, and the operating system does not allow a child to continues is its parent terminates.

Interprocess Communication

- Process executing concurrently in the operating system my be either
- Independent process or cooperating processes.

Independet processes - They cannot affect or be affect or be affected by the other processes executing in the system.

Cooperating processes – They can affect or be affected by the other processes executing in the system

- Any process that shares data with other processes in a cooperating process.

The are several reasons for providing an environment that allows process cooperation :

Information sharing

Computation speedup

Modularity

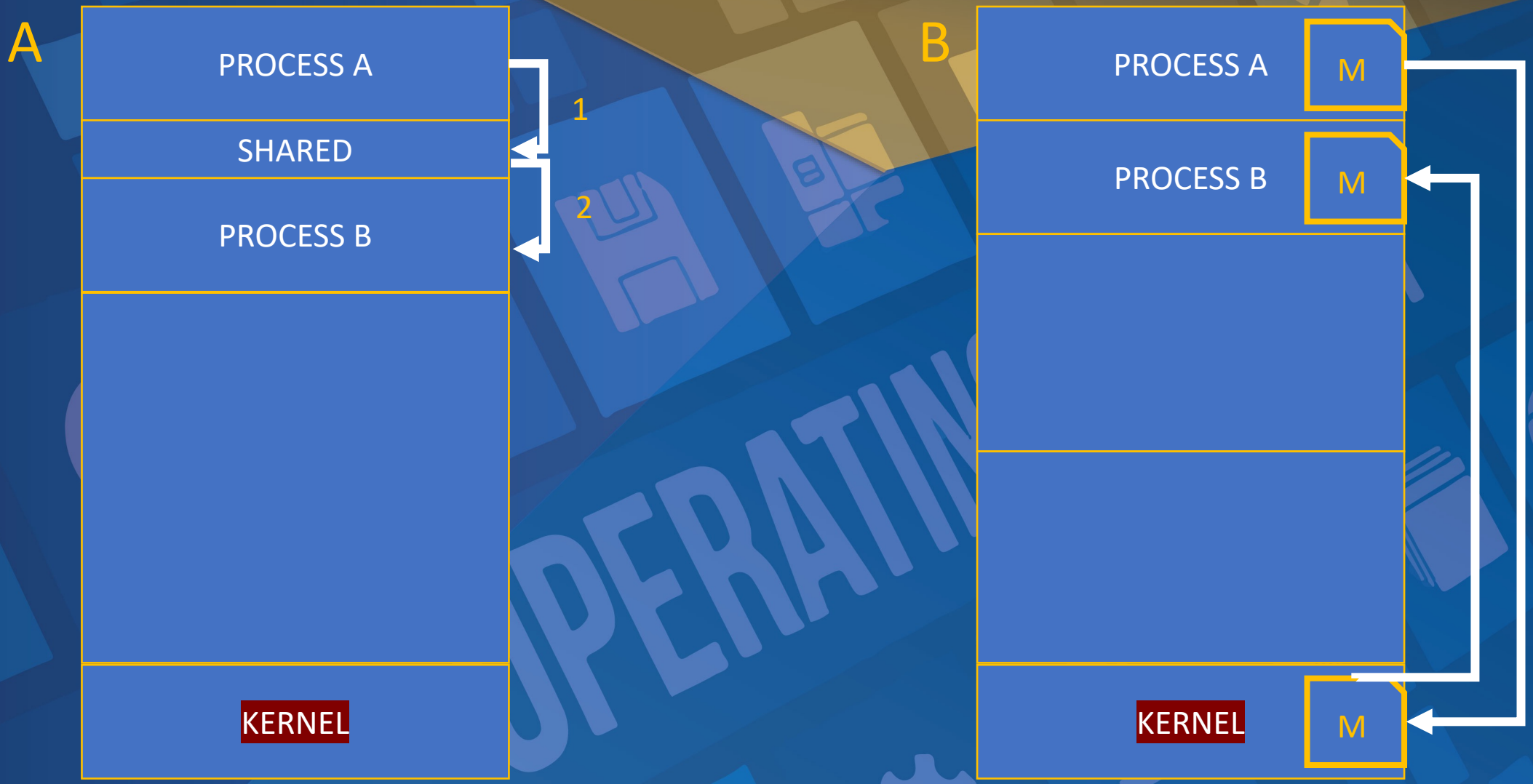
Convenience

Cooperating processes require an interprocess communication (IPC) mechanisme that will allow them to exchange data and information.

There are two fundamental models of interprocess communication:

- Shared memory
- Message passing

- In the shared-memory model, a region of memory that is shared by cooperating processes is established.
- Processes can then exchange information by reading and writing data to the shared region.
- In the message passing model, communication takes place by means of message exchanged between the cooperating processes.



COMMUNICATIONS MODELS

A. SHARED MEMORY
B. MESSAGE PASSING